BEE 4750 Homework 5: Mixed Integer and Stochastic Programming

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Due Date

Thursday, 12/04/24, 9:00pm

Overview

Instructions

- In Problem 1, you will use mixed integer programming to solve a waste load allocation problem.
- In Problem 2, you will formulate a stochastic optimization problem.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
  Activating project at `~/hw5-claire_hw5`
  Installing known registries into `~/.julia`
       Added `General` registry to ~/.julia/registries
   Installed x265_jll             v4.1.0+0
```

```
Installed libdecor_jll           v0.2.2+0
Installed JpegTurbo_jll          v3.1.3+0
Installed libfdk_aac_jll         v2.0.4+0
Installed Libmount_jll           v2.41.2+0
Installed GR_jll               v0.73.18+0
Installed OffsetArrays           v1.17.0
Installed MutableArithmetics        v1.6.7
Installed LERC_jll              v4.0.1+0
Installed Opus_jll              v1.5.2+0
Installed Xorg_xkbcomp_jll         v1.4.7+0
Installed LoggingExtras          v1.2.0
Installed RelocatableFolders        v1.0.1
Installed Preferences           v1.5.0
Installed NetworkLayout          v0.4.10
Installed InlineStrings          v1.4.5
Installed Measures              v0.3.3
Installed ConcurrentUtilities       v2.5.0
Installed Contour              v0.6.3
Installed StaticArrays           v1.9.15
Installed EarCut_jll            v2.2.4+0
Installed Grisu               v1.0.2
Installed Xorg_xcb_util_wm_jll       v0.4.2+0
Installed Xorg_xcb_util_image_jll     v0.4.1+0
Installed RecipesPipeline         v0.6.12
Installed PlotUtils             v1.4.4
Installed ColorSchemes           v3.31.0
Installed DelimitedFiles          v1.9.1
Installed Xorg_libSM_jll          v1.2.6+0
Installed Cairo_jll             v1.18.5+0
Installed InvertedIndices         v1.3.1
Installed OpenSSL              v1.6.0
Installed HTTP               v1.10.19
Installed Xorg_xcb_util_jll        v0.4.1+0
Installed OpenBLAS32_jll          v0.3.29+0
Installed Fontconfig_jll          v2.17.1+0
Installed Xorg_libxkbfile_jll       v1.1.3+0
Installed Statistics            v1.11.1
Installed Xorg_libXinerama_jll       v1.1.6+0
Installed DataFrames            v1.8.1
Installed HiGHS_jll             v1.12.0+0
Installed EpollShim_jll          v0.0.20230411+1
Installed Missings             v1.2.0
Installed FFMPEG              v0.4.5
```

```
Installed IrrationalConstants       v0.2.6
Installed Showoff             v1.0.3
Installed ArnoldiMethod          v0.4.0
Installed JSON              v1.3.0
Installed Pango_jll            v1.57.0+0
Installed PtrArrays            v1.3.0
Installed Xorg_libXau_jll         v1.0.13+0
Installed SpecialFunctions         v2.6.1
Installed Xorg_xcb_util_keysyms_jll    v0.4.1+0
Installed Bzip2_jll            v1.0.9+0
Installed BenchmarkTools          v1.6.3
Installed MathOptInterface         v1.46.0
Installed xkbcommon_jll          v1.9.2+0
Installed CodecBzip2            v0.8.5
Installed SentinelArrays          v1.4.8
Installed SimpleBufferStream        v1.2.0
Installed XZ_jll              v5.8.1+0
Installed HarfBuzz_jll           v8.5.1+0
Installed PlotThemes            v3.3.0
Installed LZO_jll             v2.10.3+0
Installed StaticArraysCore         v1.4.4
Installed NaNMath             v1.1.3
Installed TranscodingStreams        v0.11.3
Installed fzf_jll             v0.61.1+0
Installed FriBidi_jll           v1.0.17+0
Installed UnicodeFun            v0.4.1
Installed MbedTLS             v1.1.9
Installed GLFW_jll             v3.4.0+2
Installed Ratios              v0.4.5
Installed x264_jll             v10164.0.1+0
Installed CommonSubexpressions       v0.3.1
Installed Graphs              v1.13.1
Installed DisplayAs            v0.1.6
Installed Colors              v0.13.1
Installed FreeType2_jll          v2.13.4+0
Installed JLFzf              v0.1.11
Installed StatsAPI             v1.7.1
Installed DataStructures          v0.19.3
Installed GraphRecipes           v0.5.15
Installed Compat              v4.18.1
Installed METIS_jll            v5.1.3+0
Installed CodecZlib            v0.7.8
Installed Xorg_libxcb_jll         v1.17.1+0
```

```
Installed StatsBase            v0.34.8
Installed JuMP               v1.29.3
Installed libpng_jll            v1.6.50+0
Installed StructTypes           v1.11.0
Installed libaom_jll            v3.13.1+0
Installed mtdev_jll            v1.1.7+0
Installed GR                v0.73.18
Installed Dbus_jll             v1.16.2+0
Installed ExceptionUnwrapping       v0.1.11
Installed Scratch             v1.3.0
Installed DiffRules            v1.15.1
Installed eudev_jll            v3.2.14+0
Installed Xorg_libXext_jll         v1.3.7+0
Installed ArgCheck             v2.5.0
Installed ColorTypes            v0.11.5
Installed GeometryTypes          v0.8.5
Installed JSON3              v1.14.3
Installed Plots              v1.41.1
Installed TensorCore            v0.1.1
Installed Zstd_jll             v1.5.7+1
Installed Expat_jll            v2.7.3+0
Installed Xorg_xcb_util_cursor_jll     v0.1.6+0
Installed TableTraits           v1.0.1
Installed Inflate             v0.1.5
Installed DiffResults           v1.1.0
Installed Parsers             v2.8.3
Installed Libtiff_jll           v4.7.2+0
Installed PooledArrays          v1.4.3
Installed Extents             v0.1.6
Installed Format             v1.3.7
Installed JLLWrappers           v1.7.1
Installed DataValueInterfaces       v1.0.0
Installed Libffi_jll            v3.4.7+0
Installed libevdev_jll          v1.13.4+0
Installed ColorVectorSpace        v0.10.0
Installed Xorg_libXrender_jll       v0.9.12+0
Installed OrderedCollections       v1.8.1
Installed AbstractTrees          v0.4.5
Installed PrettyTables          v3.1.0
Installed ChainRulesCore         v1.26.0
Installed libinput_jll          v1.28.1+0
Installed Xorg_libXi_jll         v1.8.3+0
Installed Qt6ShaderTools_jll       v6.8.2+1
```

```
Installed Vulkan_Loader_jll         v1.3.243+0
Installed Ogg_jll              v1.3.6+0
Installed Reexport             v1.2.2
Installed Ghostscript_jll         v9.55.1+0
Installed Interpolations          v0.16.2
Installed Qt6Declarative_jll        v6.8.2+1
Installed OpenSpecFun_jll         v0.5.6+0
Installed LogExpFunctions         v0.3.29
Installed AliasTables           v1.1.3
Installed Xorg_libXcursor_jll       v1.2.4+0
Installed Libuuid_jll           v2.41.2+0
Installed HiGHS              v1.20.1
Installed MacroTools            v0.5.16
Installed SimpleTraits           v0.9.5
Installed Xorg_xcb_util_renderutil_jll   v0.3.10+0
Installed Adapt              v4.4.0
Installed DocStringExtensions       v0.9.5
Installed Graphite2_jll          v1.3.15+0
Installed Xorg_libICE_jll         v1.1.2+0
Installed Crayons             v4.1.1
Installed StableRNGs            v1.0.4
Installed libass_jll            v0.17.4+0
Installed Pixman_jll            v0.44.2+0
Installed Latexify             v0.16.10
Installed Xorg_xtrans_jll         v1.6.0+0
Installed Wayland_jll           v1.24.0+0
Installed OpenSSL_jll           v3.5.4+0
Installed BitFlags             v0.1.9
Installed IterTools            v1.10.0
Installed GeometryBasics          v0.5.10
Installed FFMPEG_jll            v8.0.0+0
Installed Xorg_xkeyboard_config_jll    v2.44.0+0
Installed LLVMOpenMP_jll          v18.1.8+0
Installed DataAPI             v1.16.0
Installed FixedPointNumbers        v0.8.5
Installed RecipesBase           v1.3.4
Installed Xorg_libXfixes_jll        v6.0.2+0
Installed LAME_jll             v3.100.3+0
Installed GettextRuntime_jll        v0.22.4+0
Installed Qt6Wayland_jll          v6.8.2+2
Installed Tables              v1.12.1
Installed PrecompileTools         v1.2.1
Installed Libiconv_jll           v1.18.0+0
```

```
   Installed Xorg_libXrandr_jll        v1.5.5+0
   Installed IteratorInterfaceExtensions   v1.0.0
   Installed Qt6Base_jll           v6.8.2+2
   Installed Glib_jll             v2.86.0+0
   Installed LaTeXStrings           v1.4.0
   Installed URIs                v1.6.1
   Installed libvorbis_jll          v1.3.8+0
   Installed Libglvnd_jll           v1.7.1+1
   Installed MathOptIIS            v0.1.1
   Installed ForwardDiff           v1.3.0
   Installed AxisAlgorithms          v1.1.0
   Installed Requires              v1.3.1
   Installed Xorg_libX11_jll         v1.8.12+0
   Installed Xorg_libXdmcp_jll        v1.1.6+0
   Installed Unzip                v0.2.0
   Installed WoodburyMatrices         v1.0.0
   Installed SortingAlgorithms        v1.2.2
   Installed StructUtils           v2.6.0
   Installed StringManipulation        v0.4.1
   Installed MarkdownTables          v1.1.0
Precompiling project...
    325.3 ms   LaTeXStrings
    279.0 ms   IteratorInterfaceExtensions
    284.7 ms   StatsAPI
    306.3 ms   TensorCore
    316.1 ms   Contour
    289.2 ms   Measures
    281.3 ms   ArgCheck
    378.3 ms   Statistics
    678.4 ms   OffsetArrays
    390.4 ms   IterTools
    743.7 ms   Format
    541.6 ms   Grisu
    235.6 ms   DataValueInterfaces
    311.0 ms   Requires
    411.7 ms   OrderedCollections
    250.4 ms   Reexport
    315.5 ms   StableRNGs
    382.6 ms   Unzip
    387.5 ms   DocStringExtensions
    375.7 ms   AbstractTrees
    488.7 ms   InlineStrings
   1525.2 ms   MacroTools
```

```
  250.2 ms  SimpleBufferStream
  456.4 ms  URIs
  335.4 ms  Extents
  236.9 ms  PtrArrays
  765.5 ms  IrrationalConstants
  372.2 ms  TranscodingStreams
  323.4 ms  NaNMath
  270.9 ms  InvertedIndices
  530.2 ms  DisplayAs
  338.5 ms  DelimitedFiles
  338.9 ms  Inflate
  280.8 ms  DataAPI
  262.8 ms  BitFlags
  308.5 ms  StaticArraysCore
  524.9 ms  ConcurrentUtilities
  775.9 ms  Crayons
  512.5 ms  WoodburyMatrices
  575.3 ms  StructTypes
  323.0 ms  Scratch
  888.0 ms  SentinelArrays
  367.1 ms  LoggingExtras
  517.0 ms  StructUtils
  738.4 ms  MbedTLS
  467.6 ms  Preferences
  414.9 ms  Compat
  357.5 ms  ExceptionUnwrapping
  393.0 ms  TableTraits
  875.4 ms  UnicodeFun
  562.0 ms  Statistics → SparseArraysExt
  277.9 ms  Showoff
  298.0 ms  Ratios
  344.9 ms  Adapt
  839.7 ms  CommonSubexpressions
 1823.4 ms  FixedPointNumbers
  885.6 ms  SimpleTraits
 1373.7 ms  DataStructures
  767.5 ms  AliasTables
  881.0 ms  CodecZlib
  356.6 ms  Missings
  810.5 ms  PooledArrays
 1296.5 ms  LogExpFunctions
  445.9 ms  DiffResults
  409.6 ms  RelocatableFolders
```

```
3387.5 ms   MutableArithmetics
 721.9 ms   AxisAlgorithms
 492.4 ms   PrecompileTools
 565.6 ms   JLLWrappers
 508.9 ms   Compat → CompatLinearAlgebraExt
 628.2 ms   Adapt → AdaptSparseArraysExt
 550.3 ms   OffsetArrays → OffsetArraysAdaptExt
1023.7 ms   Ratios → RatiosFixedPointNumbersExt
1630.9 ms   Tables
 814.5 ms   SortingAlgorithms
2163.1 ms   ColorTypes
1103.1 ms   RecipesBase
1336.5 ms   StringManipulation
 526.8 ms   OpenSSL_jll
 482.0 ms   Graphite2_jll
 355.1 ms   Libmount_jll
 333.0 ms   EpollShim_jll
 394.0 ms   LLVMOpenMP_jll
 355.8 ms   Bzip2_jll
 344.0 ms   Xorg_libICE_jll
 408.8 ms   METIS_jll
 346.1 ms   Xorg_libXau_jll
 364.7 ms   libfdk_aac_jll
 398.4 ms   libpng_jll
 406.1 ms   LAME_jll
 380.3 ms   LERC_jll
 385.8 ms   fzf_jll
 370.7 ms   EarCut_jll
 415.3 ms   JpegTurbo_jll
 422.2 ms   XZ_jll
 385.5 ms   Ogg_jll
 380.0 ms   mtdev_jll
 355.9 ms   Xorg_libXdmcp_jll
 390.5 ms   x265_jll
 412.8 ms   x264_jll
 422.8 ms   libaom_jll
 422.6 ms   Zstd_jll
 397.6 ms   Expat_jll
 367.5 ms   LZO_jll
 366.0 ms   Opus_jll
5550.4 ms   StaticArrays
 319.0 ms   libevdev_jll
 342.6 ms   Xorg_xtrans_jll
```

```
 382.3 ms    Libiconv_jll
 321.5 ms    eudev_jll
 354.8 ms    Libffi_jll
 419.7 ms    OpenSpecFun_jll
 342.1 ms    Libuuid_jll
 380.2 ms    OpenBLAS32_jll
 394.7 ms    FriBidi_jll
 827.8 ms    ChainRulesCore
 578.7 ms    StructUtils → StructUtilsTablesExt
1276.2 ms    MarkdownTables
7721.3 ms    Parsers
1978.6 ms    ColorVectorSpace
3146.5 ms    StatsBase
 889.4 ms    Pixman_jll
2938.5 ms    Colors
 593.6 ms    FreeType2_jll
 358.1 ms    CodecBzip2
 320.3 ms    Xorg_libSM_jll
1893.0 ms    OpenSSL
 588.2 ms    JLFzf
 302.4 ms    Xorg_libxcb_jll
 825.2 ms    libvorbis_jll
 322.9 ms    Dbus_jll
1195.2 ms    Ghostscript_jll
 640.2 ms    StaticArrays → StaticArraysStatisticsExt
 418.3 ms    Adapt → AdaptStaticArraysExt
1018.0 ms    ArnoldiMethod
1366.1 ms    Libtiff_jll
 300.0 ms    libinput_jll
 294.8 ms    Wayland_jll
 919.8 ms    GettextRuntime_jll
 709.8 ms    ChainRulesCore → ChainRulesCoreSparseArraysExt
 694.2 ms    LogExpFunctions → LogExpFunctionsChainRulesCoreExt
1862.9 ms    SpecialFunctions
2157.4 ms    HiGHS_jll
 495.0 ms    StaticArrays → StaticArraysChainRulesCoreExt
 457.3 ms    InlineStrings → ParsersExt
2119.8 ms    JSON
1078.5 ms    Fontconfig_jll
3176.4 ms    ColorSchemes
 376.8 ms    Xorg_xcb_util_jll
 325.5 ms    Xorg_libX11_jll
4877.4 ms    JSON3
```

```
 7975.8 ms   GeometryBasics
 1507.4 ms   GeometryTypes
 2652.0 ms   Latexify
 1126.8 ms   Glib_jll
  444.4 ms   DiffRules
  422.8 ms   ColorVectorSpace → SpecialFunctionsExt
 1166.2 ms   SpecialFunctions → SpecialFunctionsChainRulesCoreExt
 3158.4 ms   Graphs
 1045.6 ms   BenchmarkTools
  354.6 ms   Xorg_xcb_util_image_jll
  337.9 ms   Xorg_xcb_util_keysyms_jll
  358.6 ms   Xorg_xcb_util_renderutil_jll
 2282.3 ms   Interpolations
  319.4 ms   Xorg_xcb_util_wm_jll
  344.2 ms   Xorg_libXrender_jll
  345.9 ms   Xorg_libXext_jll
  376.9 ms   Xorg_libXfixes_jll
  355.8 ms   Xorg_libxkbfile_jll
  669.9 ms   Latexify → SparseArraysExt
 1216.7 ms   NetworkLayout
  348.3 ms   Xorg_xcb_util_cursor_jll
  333.7 ms   Xorg_libXinerama_jll
  356.0 ms   Xorg_libXrandr_jll
  375.4 ms   Libglvnd_jll
 3024.1 ms   ForwardDiff
22145.6 ms   PrettyTables
  357.4 ms   Xorg_libXcursor_jll
  365.3 ms   Xorg_libXi_jll
 7253.8 ms   PlotUtils
  347.7 ms   Xorg_xkbcomp_jll
 2067.9 ms   Cairo_jll
13091.2 ms   HTTP
  723.2 ms   ForwardDiff → ForwardDiffStaticArraysExt
  909.2 ms   Interpolations → InterpolationsForwardDiffExt
 1260.5 ms   NetworkLayout → NetworkLayoutGraphsExt
  902.0 ms   Xorg_xkeyboard_config_jll
 1769.6 ms   PlotThemes
 2446.7 ms   RecipesPipeline
 1606.5 ms   HarfBuzz_jll
  312.0 ms   xkbcommon_jll
 1182.5 ms   libass_jll
 1080.0 ms   Pango_jll
  332.1 ms   Vulkan_Loader_jll
```

```
   308.3 ms   libdecor_jll
  3546.9 ms   GraphRecipes
   333.0 ms   GLFW_jll
  3625.1 ms   Qt6Base_jll
  4332.8 ms   FFMPEG_jll
  2859.1 ms   FFMPEG
  3736.0 ms   Qt6ShaderTools_jll
  3856.1 ms   GR_jll
  1150.5 ms   Qt6Declarative_jll
   244.8 ms   Qt6Wayland_jll
  2184.1 ms   GR
 23879.8 ms   MathOptInterface
  1473.6 ms   MathOptIIS
 27232.6 ms   DataFrames
  1644.6 ms   Latexify → DataFramesExt
  6166.2 ms   HiGHS
 11214.9 ms   JuMP
 34174.7 ms   Plots
  2151.8 ms   Plots → GeometryBasicsExt
213 dependencies successfully precompiled in 92 seconds. 39 already precompiled.
```

```julia
using JuMP
using HiGHS
using DataFrames
using GraphRecipes
using Plots
using Measures
using MarkdownTables
```

Problems (Total: 30 Points)

Problem 1 (24 points)

Three cities are developing a coordinated municipal solid waste (MSW) disposal plan. Three disposal alternatives are being considered: a landfill (LF), a materials recycling facility (MRF), and a waste-to-energy facility (WTE). The capacities of these facilities and the fees for operation and disposal are provided below.

- **LF**: Capacity 200 Mg, fixed cost \$2000/day, tipping cost \$50/Mg;
- **MRF**: Capacity 350 Mg, fixed cost \$1500/day, tipping cost \$7/Mg, recycling cost \$40/Mg recycled;
- **WTE**: Capacity 210 Mg, fixed cost \$2500/day, tipping cost \$60/Mg;

The MRF recycling rate is 40%, and the ash fraction of non-recycled waste is 16% and of recycled waste is 14%. Transportation costs are $1.5/Mg-km, and the relative distances between the cities and facilities are provided in the table below.

City/Facility	Landfill (km)	MRF (km)	WTE (km)
1	5	30	15
2	15	25	10
3	13	45	20
LF	-	32	18
MRF	32	-	15
WTE	18	15	-

The fixed costs associated with the disposal options are incurred only if the particular disposal option is implemented. The three cities produce 100, 90, and 120 Mg/day of solid waste, respectively, with the composition provided in the table below.

Component	% of total mass	Combustion ash (%)	MRF Recycling rate (%)
Food Wastes	15	8	0
Paper & Cardboard	40	7	55
Plastics	5	5	15
Textiles	3	10	10
Rubber, Leather	2	15	0
Wood	5	2	30
Yard Wastes	18	2	40
Glass	4	100	60
Ferrous	2	100	75
Aluminum	2	100	80
Other Metal	1	100	50
Miscellaneous	3	70	0

The information in the above table will help you determine the overall recycling and ash fractions. Note that the recycling residuals, which may be sent to either landfill or the WTE, have different ash content than the ash content of the original MSW. You will need to determine these fractions to construct your mass balance constraints.

Reminder: Use `round(x; digits=n)` to report values to the appropriate precision!

Problem 1.1

Based on the information above, calculate the overall recycling and ash fractions for the waste produced by each city.

Problem 1.2

What are the decision variables for your optimization problem? Provide notation and variable meaning.

Problem 1.3

Formulate the objective function. Make sure to include any needed derivations or justifications for your equation(s).

Problem 1.4

Derive all relevant constraints. Make sure to include any needed justifications or derivations.

Problem 1.5

Find the optimal solution (using `JuMP` to solve the problem). Report the optimal objective value.

Problem 1.6

Draw a diagram showing the flows of waste between the cities and the facilities. Which facilities (if any) will not be used? Does this solution make sense?

Problem 2 (6 points)

Consider a two-period economic dispatch problem, based on the multi-period example from Lecture 14 (on 10/29). The generator data, including ramping constraints for each generator, is provided in 'data/generators.csv.' In period 1, the demand is $d_1 = 1100$MW. In period 2, the demand is projected to be $d_2 = 1200$MW, but there is a 25% probability that it is \$1500 . In the first period, the solar capacity factor is 0.9 and the wind capacity factor is 0.45, but in the second period, there is some uncertainty: the forecasted solar and wind capacity factors are 0.95 and 0.4, respectively, but there is a 30% probability that they are 0.75 and 0.5. Your goal is to identify how to dispatch your generators to minimize the cost of meeting demand.

Problem 2.1

Draw a scenario tree for this problem.

Problem 2.2

Formulate a stochastic linear program for this problem based on your scenario tree from Problem 2.1 and the data in `data/generators.csv`.

References

List any external references consulted, including classmates.